

Economic Interpretations of LP Duality

- A company produces three products A, B, C with two different resources I and II

Material uses & profits:

	Product A	Product B	Product C
Resource I	a_{11}	a_{12}	a_{13}
Resource II	a_{21}	a_{22}	a_{23}
Profit	C_1	C_2	C_3

Limitation of Resources I b_1

II: b_2

- The company wants to maximize its profit with the current resources.

$$\begin{aligned} \max \quad & C_1 X_1 + C_2 X_2 + C_3 X_3 \\ \text{s.t.} \quad & a_{11} X_1 + a_{12} X_2 + a_{13} X_3 \leq b_1 \\ & a_{21} X_1 + a_{22} X_2 + a_{23} X_3 \leq b_2 \\ & X_1, X_2, X_3 \geq 0 \end{aligned} \quad (P)$$

On the other hand, there are potential buyers of resources. If the prices of resources are sufficiently high, the company will sell the resources instead of selling the products.

- Suppose that the prices of the resources are $\pi_1 \geq 0$ and $\pi_2 \geq 0$.

If the company reduces the production of Product A by 1 unit, then a_{11} units of resource I and a_{21} units of resource II will be released. The company will lose C_1 in profit, but will make $a_{11}\pi_1 + a_{21}\pi_2$ by selling the released resources. From the viewpoint of the company, it will sell its resources only if

$$a_{11}\pi_1 + a_{21}\pi_2 \geq C_1$$

Similarly, decreasing the production of Products B and C would result in the following constraints.

$$a_{12}\pi_1 + a_{22}\pi_2 \geq C_2$$

$$a_{13}\pi_1 + a_{23}\pi_2 \geq C_3$$

- Consider the buyer's point of view. The total price of the resources is $b_1\pi_1 + b_2\pi_2$, and the buyer wants to minimize the total price. Therefore, both parties will be satisfied if π_1^* and π_2^*

solves:

$$\min Z = b_1\pi_1 + b_2\pi_2$$

$$\text{S.t. } a_{11}\pi_1 + a_{21}\pi_2 \geq C_1$$

$$a_{12}\pi_1 + a_{22}\pi_2 \geq C_2 \quad (D)$$

$$a_{13}\pi_1 + a_{23}\pi_2 \geq C_3$$

$$\pi_1, \pi_2 \geq 0$$

As we can see, the dual problem can be thought of as the problem of determining fair market prices of the resources. The dual variables π_1^*, π_2^* are called the shadow prices of the resources

- Complementary slackness. $\pi_i^* (b - Ax^*)_i = 0$

If Resource i is not used up $((b - Ax^*)_i > 0)$, then its shadow price is zero ($\pi_i^* = 0$).

If the shadow price of Resource i is nonzero ($\pi_i^* > 0$), then the resource is used up $((b - Ax^*)_i = 0)$.